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Electric motor including windings having crossovers

Abstract

A method of arranging windings in a parallel delta configuration in a stator assembly includes connecting a first lead of a first winding to a first terminal. The first terminal is electrically connected to a printed circuit board assembly (PCBA) located at a first axial end of the stator assembly. The method also includes wrapping the first winding about a first tooth of a lamination stack to form a first coil, extending the first winding about at least a portion of a circumference of the lamination stack at a second axial end of the stator assembly toward a second tooth located opposite the first tooth, connecting the first winding to a second terminal electrically connected to the PCBA, wrapping the first winding about the second tooth of the lamination stack to form a second coil, and connecting a second lead of the first winding to the first terminal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 18/057,844 filed on Nov. 22, 2022, now U.S. Pat. No. 12,003,137, which is a continuation of U.S. patent application Ser. No. 16/698,293 filed on Nov. 27, 2019, now U.S. Pat. No. 11,515,743, which claims priority to U.S. Provisional Patent Application No. 62/772,934 filed on Nov. 29, 2018, the entire contents of each of which is incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to tools, such as power tools, and more particularly to power tools including an electric motor having windings wound on a stator.

BACKGROUND OF THE INVENTION

(2) Tools, such as power tools, can include an electric motor having a stator to generate a magnetic field. The stator may include windings wound in slots to form coils, and terminals electrically connecting the coils to a power source.

SUMMARY OF THE INVENTION

(3) The present invention includes, in one aspect, a method of arranging a plurality of windings in a

parallel delta configuration in a stator assembly. The method includes connecting a first lead of a first winding of the plurality of windings to a first terminal. The first terminal is electrically connected to a printed circuit board assembly (PCBA) located at a first axial end of the stator assembly. The method also includes wrapping the first winding about a first tooth of a lamination stack to form a first coil. The method further includes extending the first winding about at least a portion of a circumference of the lamination stack at a second axial end of the stator assembly toward a second tooth located opposite the first tooth. The method also includes connecting the first winding to a second terminal electrically connected to the PCBA. The method further includes wrapping the first winding about the second tooth of the lamination stack to form a second coil. The method also includes connecting a second lead of the first winding to the first terminal.

(4) The present invention includes, in another aspect, an electric motor including a stator assembly having a lamination stack with an annular yoke and a plurality of teeth extending inwardly from the yoke. The teeth define a plurality of slots therebetween. The plurality of slots include a plurality of first slots and a plurality of second slots. The first slots and the second slots are formed in an alternating sequence in a circumferential direction of the stator assembly. The stator assembly also includes a printed circuit board assembly (PCBA) coupled to the lamination stack. The stator assembly further includes a plurality of terminals electrically connected to the PCBA. The stator assembly also includes a plurality of windings wrapped about the lamination stack to form coils. Each winding includes crossover portions extending about a portion of a circumference of the stator assembly to connect pairs of opposite coils. The windings include leads that contact the terminals to electrically connect the coils to the PCBA. The leads extend longitudinally through the first slots only to connect the terminals to the crossover portions.

(5) Other features and aspects of the invention will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front view of an electric motor according to an embodiment of the invention.
- (2) FIG. 2 is a schematic cross-sectional view of a stator assembly of the electric motor of FIG. 1.
- (3) FIG. 3 is a schematic diagram of a prior art winding configuration.
- (4) FIG. 4 is a schematic diagram of a winding configuration of the electric motor of FIG. 1.
- (5) FIG. 5 is a detail view of a portion of the electric motor of FIG. 1.
- (6) FIG. 6 is another detail view of a portion of the electric motor of FIG. 1.
- (7) FIG. 7 is a schematic diagram of the winding configuration of FIG. 4.
- (8) FIG. 8 is a front view of an electric motor according to another embodiment of the invention.
- (9) Before any embodiments of the application are explained in detail, it is to be understood that the application is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The application is capable of other embodiments and of being practiced or of being carried out in various ways.

DETAILED DESCRIPTION

- (10) FIGS. 1, 2, and 4-7 illustrate an electric motor **100** for use in various different tools, such as power tools (e.g., rotary hammers, pipe threaders, cutting tools, etc.), outdoor tools (e.g., trimmers, pole saws, blowers, etc.), and other electrical devices (e.g., motorized devices, etc.).
- (11) The electric motor **100** is configured as a brushless DC motor. In some embodiments, the motor **100** may receive power from an on-board power source (e.g., a battery, not shown). The battery may include any of a number of different nominal voltages (e.g., 12V, 18V, etc.), and may be configured having any of a number of different chemistries (e.g., lithium-ion, nickel-cadmium,

etc.). Alternatively, the motor **100** may be powered by a remote power source (e.g., a household electrical outlet) through a power cord. The motor **100** includes a substantially cylindrical stator assembly **102** operable to produce a magnetic field, and a rotor assembly (not shown) supported for rotation with respect to the stator assembly **102**.

(12) The stator assembly **102** includes a lamination stack **106** formed from a plurality of laminations that are stacked along a rotational axis **108**. In the illustrated embodiment, six stator teeth **TT1**, **TT2** (FIG. 2) extend inwardly in a generally radial direction from an annular yoke **110** of the lamination stack **106**. Pairs of adjacent teeth **TT1**, **TT2** form slots **S1**, **S2** therebetween. Although the stator assembly **102** of the illustrated embodiment includes an annular yoke **110**, in other embodiments (not shown) the yoke may be formed in other, non-round shapes (e.g., square, hexagonal, etc.).

(13) Windings **W** (FIG. 1) are routed through the slots **S1**, **S2** (FIG. 2) and wound about the respective teeth **TT1**, **TT2** to form coils **C1**, **C2**, **C3**, **C4**, **C5**, and **C6** (FIG. 7). The windings **W** are electrically connected to terminals **T** (FIG. 1) (e.g., by direct contact), which in turn are electrically connected to a PCBA **112** (e.g., a hall effect sensor board) that selectively supplies current to the coils **C1-C6** via the terminals **T**.

(14) In some prior art designs as illustrated by FIG. 3, the windings are routed in a series configuration in which a single winding **W** extends from a first terminal to a first tooth and forms a first coil, then extends from the first coil to a second, opposite tooth and forms a second coil, and then extends from the second coil to a second terminal and attaches thereto. In the proposed electric motor **100** embodied in FIGS. 1, 2, and 4-7, the windings **W** are configured in a parallel delta configuration (FIG. 4). Specifically, each individual winding **W** extends between pairs of electrically opposite coils **C1-C4**, **C2-C5**, and **C3-C6**, respectively (FIG. 4). Each pair of opposite coils **C1-C4**, **C2-C5**, and **C3-C6** is connected in parallel between respective terminals **T**.

(15) Since the coils of each pair **C1-C4**, **C2-C5**, and **C3-C6** are located opposite one another, portions of the windings **W** extend around the circumference of the stator assembly **102** to connect the coils of each pair **C1-C4**, **C2-C5**, and **C3-C6**. These portions of the windings **W** are referred to herein as crossovers **114** (FIG. 1). In the illustrated embodiment, the crossovers **114** are routed along a fan side **116** of the electric motor **100**, and the PCBA **112** is coupled to the motor **100** at a board side **118** of the electric motor **100** opposite the fan side **116**. In the illustrated embodiment, the terminals **T** are coupled to the stator assembly **102** at the board side **118**, opposite the crossovers **114**. In some embodiments, the electric motor **100** further includes a fan (not shown) rotated by the rotor assembly (not shown), and the fan is located adjacent the fan side **116**.

(16) Each terminal **T** includes a hook portion or tang **120** (FIG. 5), and the windings **W** are wrapped around the tangs **120** to form the electrical connections between the windings **W** and the terminals **T**. This allows the winding configuration to be implemented automatically by a needle winder. The winding configuration also permits the needle winder to wind three of the coils **C1-C6** at one time, which can improve the throughput of the winding process. Referring to FIG. 5, the tang **120** extends outward from the stator assembly **102** in a radial direction, and a looped lead **122** of the windings **W** may be wound around the tang **120** to form the electrical connection between the windings **W** and the terminal **T**. In some embodiments, in addition to the tang **120**, a second tang **120'** may extend away from the terminal **T** in a circumferential direction as illustrated in FIG. 6.

(17) The windings **W** may include a first winding **W1**, a second winding **W2**, and a third winding **W3**, each wound about the stator assembly **102** in the configuration illustrated by FIG. 7. Likewise, the terminals **T** may include a first terminal **T1**, a second terminal **T2**, and a third terminal **T3**. Each of the windings **W1**, **W2**, and **W3** includes a respective start lead **124** and a respective finish lead **126** that each connect to a respective terminal **T1**, **T2**, or **T3**.

(18) For example, as shown in FIG. 7, the first winding **W1** begins at the start lead **124** at the second terminal **T2**, passes longitudinally through the slot **S1**, and wraps about the stator assembly

102 to form the first coil **C1**. The first winding **W1** exits the first coil **C1** at the fan side **116** and extends about the circumference of the stator assembly **102** (via the crossover **114**) toward the location of fourth coil **C4**. The first winding **W1** then extends longitudinally through the slot **S1** adjacent the fourth coil **C4** to form the looped lead **122** about the tang **120** (FIG. 5) of the first terminal **T1**, thereby electrically connecting the first winding **W1** to the first terminal **T1**. The first winding **W1** then reverses course, passing back through the slot **S1** in the longitudinal direction, and wrapping around the stator assembly **102** to form the fourth coil **C4**. The first winding **W1** exits the fourth coil **C4** on the fan side **116**, and travels back around the circumference of the stator assembly **102** (via the crossover **114**) toward the first coil **C1**. The first winding **W1** then passes longitudinally through the slot **S1** adjacent the first coil **C1**, and terminates at the finish lead **126**, which connects to the terminal **T2**. In this way, the first and fourth coils **C1**, **C4** are connected in parallel between the first and second terminals **T1**, **T2**. The second and third windings **W2**, **W3** are wound in a manner similar to the configuration described with respect to **W1**, as illustrated in FIG. 7.

(19) With continued reference to FIG. 7, the start and finish leads **124**, **126** and the looped leads **122** for each winding **W1**, **W2**, and **W3** all pass longitudinally through slots **S1** between the fan side **116** and the board side **118**. As a result, each of the slots **S1** contains more conductors than each of the slots **S2**. To account for these extra conductors, the lamination stack **106** is provided with the slots **S1** that are larger in area than the slots **S2**, as will be discussed in further detail below. The larger area of the slots **S1** helps to avoid uneven slot fill in the stator assembly **102**.

(20) With reference to FIG. 2, each tooth **TT1**, **TT2** of the lamination stack **106** includes a crown portion **128** and a body portion **130** that connects the crown portion **128** to the yoke **110**. The crown portions **128** are equally spaced from one another in a circumferential direction. However, the body portions **130** of the teeth **TT1**, **TT2** are alternately offset or skewed in a radial direction, so that each successive tooth **TT1**, **TT2** alternately skews toward the clockwise direction or toward the counter clockwise direction. For example, the teeth **TT1** skew slightly clockwise in the circumferential direction, while the teeth **TT2** skew slightly counter clockwise in the circumferential direction.

(21) For purposes of illustration, a dividing line **132** divides the lamination stack **106** into equal halves and extends through a center point **C** of the crown portion **128** of two opposing teeth **TT1** and **TT2**. A longitudinal axis **134** extends through the body portion **130** of the tooth **TT1** and intersects the dividing line **132**. The dividing line **132** forms an angle **A1** with the longitudinal axis **134** of tooth **TT1**. In the illustrated construction, the angle **A1** is approximately 166 degrees. Accordingly, tooth **TT1** is skewed or offset at an angle **A1** of approximately 166 degrees toward the clockwise direction. Likewise, each of the remaining teeth **TT1** are also offset at angles **A1** (not shown) of approximately 166 degrees toward the clockwise direction, while the alternating teeth **TT2** are offset at corresponding angles **A2** (not shown) of approximately 166 degrees toward the counter clockwise direction. In other embodiments, the angles **A1**, **A2** may be greater than 90 degrees and less than 180 degrees.

(22) Because the teeth **TT1**, **TT2** are alternately skewed or offset, the slots **S1**, **S2** defined between adjacent pairs of teeth **TT1**, **TT2** define two different slot areas **B1** and **B2**. The slot area **B1** is larger than the slot area **B2**. In the illustrated embodiment, slot area **B1** is approximately 70 millimeters squared (mm²), while slot area **B2** is approximately 60 mm². In other embodiments (not shown), the two slot areas **B1**, **B2** may be larger or smaller than 70 mm² and 60 mm², respectively, and may vary as a function of the angles **A1**, **A2**. The two different slot areas **B1** and **B2** help to avoid uneven slot fill when different numbers of windings **W** are alternately applied to the teeth **TT1**, **TT2**.

(23) FIG. 8 illustrates another embodiment of an electric motor **200** similar to the electric motor **100** described above, with like features shown with reference numerals plus "100," or, prime "(" in the case of alphabetical reference numerals. The electric motor **200** includes stator assembly **202**

operable to produce a magnetic field, and a rotor assembly (not shown) supported for rotation with respect to the stator assembly **202**. The stator assembly **202** likewise includes a lamination stack **206** and windings **W'** routed through the slots **S1**, **S2** to form coils **C1'**, **C2'**, **C3'**, **C4'**, **C5'**, and **C6'**. The windings **W'** are electrically connected to terminals **T'**, which in turn are electrically connected to a PCBA **212** (e.g., a hall effect sensor board) that selectively supplies current to the coils **C1'**-**C6'** via the terminals **T'**.

(24) In the illustrated embodiment, the windings **W'** are configured in a parallel delta configuration wherein each individual winding **W'** extends between pairs of electrically opposite coils **C1'**-**C4'**, **C2'**-**C5'**, and **C3'**-**C6'**, respectively. Each pair of opposite coils **C1'**-**C4'**, **C2'**-**C5'**, and **C3'**-**C6'** is connected in parallel between respective terminals **T'**. Crossovers **214** are routed along a fan side **216** of the electric motor **200**, and a PCBA **212** is coupled to the electric motor **200** at a board side **218** of the motor **200** opposite the fan side **216**.

(25) The terminals **T'** of the electric motor **200** are elongated such that the terminals **T'** extend longitudinally along an outer surface **240** of the lamination stack **206** between the board side **218** and the fan side **216**. Each elongated terminal **T'** includes a longitudinal portion **236** having a connecting portion **238** electrically coupled to the PCBA **212** at the board side **218** of the electric motor **200**. The longitudinal portion **236** extends along the outer surface **240** of the lamination stack **206** from the connecting portion **238** to a hook portion or tang **220** located adjacent the fan side **216**. Each tang **220** extends along the outer surface **240** of the lamination stack **206** toward the fan side **216**, and then bends backward toward the board side **218** at a location adjacent an axial end of the lamination stack **206**. Start and finish leads **224**, **226** of the windings **W'** connect to the terminals **T'** at the tangs **220** adjacent the fan side **216**. Similarly, looped leads **222** of the windings **W'** also connect to the tangs **220** adjacent the fan side **216**. The elongated terminals **T'** electrically connect the coils **C1'**-**C6'** to the PCBA **212**.

(26) This configuration allows for significant stack-up reduction to minimize the length of the motor **200** and reduce material and process costs, and also allows the winding configuration to be implemented automatically by a needle winder. The winding configuration also permits the needle winder to wind three of the coils **C1'**-**C6'** at one time, which can improve the throughput of the winding process. Moreover, the elongated terminals **T'** bridge the length of the stator assembly **202** between the PCBA **212** at the board side **218** and the crossovers **214** at the fan side **216**. Thus, the elongated terminals eliminate the need for the start and finish leads **224**, **226** and the looped leads **222** to pass through the slots **S1'** as described above with respect to the electric motor **100**. This reduces the overall length of the windings **W'** and thereby further reduces material costs.

(27) In the illustrated embodiment, the electric motor **200** includes three elongated terminals **T'** that are spaced apart from one another in the circumferential direction at generally equal intervals. Each terminal **T'** is attached to the outer surface **240** of the lamination stack **206** via an attachment member **242**. The stator assembly **202** also includes an insulating member **244** formed of an insulative material (e.g., plastic) that insulates the windings **W'** from the lamination stack **206**. In the illustrated embodiment, the attachment members **242** are separate elements formed separately from the insulating member **244**. In other embodiments (not shown), the attachment members **242** may be formed as portions of the insulating member **244**, so that the elongated terminals **T'** are secured to the electric motor **200** via the insulating member **244**.

(28) Although the application has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the scope and spirit of one or more independent aspects of the application as described.

(29) Various features of the invention are set forth in the following claims.

Claims

1. A method of arranging a plurality of windings in a parallel delta configuration in a stator assembly, the method comprising: connecting a first lead of a first winding of the plurality of windings to a first terminal, the first terminal being electrically connected to a printed circuit board assembly (PCBA) located at a first axial end of the stator assembly; wrapping the first winding about a first tooth of a lamination stack to form a first coil; extending the first winding about at least a portion of a circumference of the lamination stack at a second axial end of the stator assembly toward a second tooth located opposite the first tooth; connecting the first winding to a second terminal electrically connected to the PCBA; wrapping the first winding about the second tooth of the lamination stack to form a second coil; and connecting a second lead of the first winding to the first terminal.
2. The method of claim 1, wherein after connecting the first lead to the first terminal, the method comprises extending the first winding longitudinally through a first slot located adjacent the first tooth.
3. The method of claim 1, wherein prior to connecting the first winding to the second terminal, the method comprises extending the first winding longitudinally through a second slot located adjacent the second tooth.
4. The method of claim 3, wherein after connecting the first winding to the second terminal, the method comprises extending the first winding longitudinally back through the first slot.
5. The method of claim 1, wherein the first lead connects to the first terminal at the first axial end, and the second lead connects to the first terminal at the first axial end.
6. The method of claim 1, further comprising: connecting a third lead of a second winding of the plurality of windings to the second terminal; wrapping the second winding about a third tooth of the lamination stack to form a third coil; connecting the second winding to a third terminal electrically connected to the PCBA; wrapping the second winding about a fourth tooth of the lamination stack located opposite the third tooth to form a fourth coil; and connecting a fourth lead of the second winding to the second terminal.
7. The method of claim 6, further comprising: connecting a fifth lead of a third winding of the plurality of windings to the third terminal; wrapping the third winding about a fifth tooth of the lamination stack to form a fifth coil; connecting the third winding to the first terminal; wrapping the third winding about a sixth tooth of the lamination stack located opposite the fifth tooth to form a sixth coil; and connecting a sixth lead of the third winding to the third terminal.
8. An electric motor comprising: a stator assembly including a lamination stack including an annular yoke and a plurality of teeth extending inwardly from the yoke, the teeth defining a plurality of slots therebetween, the plurality of slots comprising a plurality of first slots and a plurality of second slots, the first slots and the second slots being formed in an alternating sequence in a circumferential direction of the stator assembly, a printed circuit board assembly (PCBA) coupled to the lamination stack, a plurality of terminals electrically connected to the PCBA; and a plurality of windings wrapped about the lamination stack to form coils, each winding including crossover portions extending about a portion of a circumference of the stator assembly to connect pairs of opposite coils, the windings including leads that contact the terminals to electrically connect the coils to the PCBA, the leads extending longitudinally through the first slots only to connect the terminals to the crossover portions.
9. The electric motor of claim 8, further including a fan located adjacent the crossover portions.
10. The electric motor of claim 8, wherein each terminal includes a tang, and wherein each winding includes a looped lead that wraps about the tang of at least one terminal to electrically connect the coils to the PCBA.
11. The electric motor of claim 8, wherein each tooth includes a crown portion and a body portion connecting the crown portion to the yoke, and wherein the body portion of each tooth is skewed away from a radial direction in a circumferential direction, and wherein the body portions of two

adjacent teeth of the plurality of teeth alternately skew toward and away from one another in the circumferential direction.

12. The electric motor of claim 8, wherein each first slot has a first slot area, and each second slot has a second slot area, the first slot area being larger than the second slot area.

13. The electric motor of claim 8, wherein the stator assembly includes a first axial end and a second axial end, and wherein the PCBA is located at the first axial end and the crossover portions are located at the second axial end.
